

# IDC 410: Machine Learning

Spring Semester 2025-26

## Assignment 09

Instructor: Prof. Shashikant R. Dugad

Uploaded on 2<sup>nd</sup> April 2026 (Thursday)

Submission: Before 9<sup>th</sup> March 2026 (Thursday) 23:00 hrs

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1. Install **COCO** dataset containing **80 classes** on your laptop. Use **YOLOv8** to identify objects in this dataset using **pre-trained** model on the COCO dataset. Describe the results of your evaluation.
2. Take a short video of about 30 seconds. Write a program to extract each frame of this video and save each frame as jpg file with naming convention frame\_nnnnn.jpg. **nnnnn** indicate the frame number in the video. Describe properties of the video and the extracted jpg files.
3. a) Describe the importance of **annotation task** in ML.  
b) Install the annotation software **labelme** (using pip install command) on your laptop.  
c) Describe its features and the content stored in the output file.  
d) Do annotation of some of the objects on the extracted frames from the video that you have taken (in Problem 2) and create the csv file or a text file of annotation as the output file.

**Please NOTE:**

**a) Complete your assignment in time**

**b) If you need any clarifications, feel free to contact TA's or me well in advance so that you can complete your assignment in time**